

PROJECT DOCUMENTATION

MICROSOFT SENTINEL BASED SOC MONITORING AND FAILED LOGON DETECTION

A Hands-on Cyber Security Monitoring Project

Submitted To	TechByHeart Academy, Kochi
Course	Advanced Diploma in Cyber Security
Project Area	Security Operations Center (SOC) / SIEM
Student Name	_____
Register No.	_____
Academic Year	_____

Submitted as part of the practical project work for the Advanced Diploma in Cyber Security.

ABSTRACT

This project demonstrates a practical Security Operations Center (SOC) monitoring workflow using Microsoft Sentinel. An isolated virtual lab was created with Kali Linux and Windows 10 virtual machines using VirtualBox. Controlled failed SMB authentication attempts were generated from Kali Linux against the Windows virtual machine. These attempts produced Windows Security Event ID 4625, which was used as the primary security telemetry. Azure Arc and Azure Monitor Agent (AMA) were used for endpoint onboarding and telemetry collection, while a Data Collection Rule (DCR) defined the Windows Security events to be collected. Microsoft Sentinel and Kusto Query Language (KQL) were used to query and detect repeated failed logon activity. A scheduled Analytics Rule was configured to identify three or more failures from the same computer and source IP within a five-minute window.

TABLE OF CONTENTS

1. Introduction
2. Problem Statement
3. Objectives
4. Existing System
5. Proposed System
6. System Requirements
7. Tools and Technologies
8. System Architecture
9. Methodology / Implementation
10. Windows Event ID 4625
11. Kali Linux Attack Simulation
12. Azure Arc and Azure Monitor Agent
13. Data Collection Rule
14. Microsoft Sentinel and KQL
15. Analytics Rule
16. Results and Analysis
17. Limitations
18. Future Scope
19. Conclusion
20. References

1. Introduction

A Security Operations Center (SOC) continuously monitors security telemetry to identify suspicious activity and support investigation. A Security Information and Event Management (SIEM) platform helps analysts collect, search, correlate, and detect events from monitored systems. This project uses Microsoft Sentinel to demonstrate a small-scale SOC monitoring workflow in a controlled virtual environment.

2. Problem Statement

Repeated failed authentication attempts can indicate password guessing, brute-force activity, or other unauthorized access attempts. Manually checking Windows Security logs is time-consuming and does not provide an efficient centralized detection workflow. This project addresses the problem by collecting Windows failed-logon events and applying KQL-based detection logic.

3. Objectives

The main objectives are to build an isolated SOC lab, generate controlled authentication failures, understand Windows Event ID 4625, onboard a Windows endpoint using Azure Arc, deploy Azure Monitor Agent, configure a Data Collection Rule, query security telemetry using KQL, and create a Microsoft Sentinel scheduled Analytics Rule.

4. Existing System

In a basic endpoint-only setup, administrators can inspect Windows Security logs directly using Event Viewer. Although this provides detailed information, it is less convenient for centralized monitoring, historical searching, correlation, and automated detection.

5. Proposed System

The proposed system centralizes selected Windows Security events in Microsoft Sentinel. The endpoint produces Event ID 4625, Azure Monitor Agent collects the configured telemetry, the DCR controls collection, Log Analytics stores the data, and Sentinel provides KQL-based analysis and Analytics Rules.

6. System Requirements

Hardware: a Windows host capable of running VirtualBox and two virtual machines, with sufficient RAM and storage, and Internet connectivity for Azure services. Software: Oracle VirtualBox, Kali Linux, Windows 10 Pro 64-bit, Microsoft Azure, Microsoft Sentinel, Azure Arc, Azure Monitor Agent, and a Log Analytics Workspace.

7. Tools and Technologies

Microsoft Sentinel was used for SIEM monitoring; KQL for querying and detection; Azure Arc for endpoint onboarding; Azure Monitor Agent for telemetry collection; Data Collection Rules for event selection; Log Analytics for centralized storage; Windows Event Viewer for endpoint validation; Kali Linux for controlled security testing; SMB for the authentication test; and VirtualBox for virtualization.

8. System Architecture

The lab consists of two virtual machines connected through an isolated VirtualBox host-only network. Kali Linux acted as the controlled attacker/test machine and Windows 10 acted as the victim/test machine. The monitoring pipeline was: Kali Linux → controlled SMB authentication failure → Windows Event ID 4625 → Azure Monitor Agent → Data Collection Rule → Log Analytics → Microsoft Sentinel → KQL detection → Analytics Rule.

9. Methodology / Implementation

The implementation was completed in stages. The virtual machines were configured with an isolated host-only network. Windows logon failure auditing was enabled and a dedicated lab account was used. The Windows endpoint was onboarded to Azure Arc and the Azure Monitor Agent extension was deployed. A Data Collection Rule was configured to collect Event ID 4625. Sentinel queries and a scheduled Analytics Rule were then configured for repeated failed-logon detection.

10. Windows Event ID 4625

Windows Security Event ID 4625 is generated when an account fails to log on. In this project it was the primary detection event. The controlled SMB test generated multiple 4625 events. Investigation fields included the target username, source IP address, logon type, workstation name, and authentication package.

11. Kali Linux Attack Simulation

A controlled SMB authentication test was performed from Kali Linux against the Windows lab VM using an intentionally incorrect password. The test was restricted to the user's own virtual machines and was performed for security monitoring practice.

```
for i in {1..5}; do smbclient -L //192.168.56.102 -U 'socuser%WrongPass123!'; done
```

12. Azure Arc and Azure Monitor Agent

The Windows VM was onboarded to Azure Arc as DESKTOP-9LVMV9D. Azure Monitor Agent was deployed through the Azure Arc machine extension. AMA is responsible for collecting endpoint telemetry defined by the associated Data Collection Rule.

13. Data Collection Rule

A Data Collection Rule named dcr-windows-securityevents was configured to collect Windows Security Event ID 4625 and send the selected telemetry to the Log Analytics workspace soc-log-workspace. The configured event filter was `Security!*[System[(EventID=4625)]]`.

14. Microsoft Sentinel and KQL

Microsoft Sentinel was used as the SIEM platform for querying and analysing collected Windows Security telemetry. KQL was used to filter Event ID 4625 and identify repeated failures by source IP and time window.

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts = count() by Computer, IPAddress, bin(TimeGenerated, 5m)
| where FailedAttempts >= 3
| order by TimeGenerated desc
```

15. Analytics Rule

A scheduled Microsoft Sentinel Analytics Rule named SOC - Multiple Failed Logons 4625 was configured with Medium severity. It was configured to run every five minutes with a ten-minute lookback and to detect three or more failed logons from the same computer and source IP within a five-minute grouping window. Sentinel health monitoring showed successful scheduled executions.

16. Results and Analysis

The project successfully demonstrated controlled failed-logon generation, Windows Event ID 4625 recording, Azure Arc onboarding, Azure Monitor Agent deployment, DCR configuration, Sentinel event querying, KQL detection logic, and scheduled Analytics Rule execution. Final alert/incident generation was not successfully verified during the project period and is therefore not claimed as a completed result.

17. Limitations

The project was implemented as a small learning lab using virtual machines and was not intended as a production SOC environment. Telemetry collection can depend on agent state, ingestion delay, network connectivity, and cloud configuration. The final alert/incident generation stage was not verified.

18. Future Scope

The project can be extended by collecting additional Windows events and Sysmon telemetry, adding more Sentinel Analytics Rules, creating incident investigation playbooks, integrating endpoint detection and response telemetry, using SOAR for automated response, and adding additional controlled SOC scenarios.

19. Conclusion

This project provided practical experience with a cloud-based SIEM and the basic SOC monitoring lifecycle. It demonstrated how a controlled authentication failure on a Windows endpoint can become a security event, be collected through Azure monitoring components, and be queried and detected in Microsoft Sentinel using KQL. It also provided practical exposure to Azure Arc, Azure Monitor Agent, Data Collection Rules, Log Analytics, Windows Event Viewer, Kali Linux, SMB, and detection engineering.

20. References

- Microsoft Learn – Microsoft Sentinel documentation.
- Microsoft Learn – Azure Monitor Agent documentation.
- Microsoft Learn – Azure Arc-enabled servers documentation.
- Microsoft Learn – Data Collection Rules documentation.
- Microsoft Learn – Kusto Query Language (KQL) documentation.